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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 314 — Past Questions & Materials
by Renance

Course: PHY 314
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200, 300

This study pack was compiled and restructured by Renance for PHY 314 students at Federal University of Petroleum Resources, Effurun. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 5 — PHY 314 - 2016/2017 - First Semester - Solid State Physics

Level 300 • First Semester • PastQuestion

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DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATIONS 2016/2017 SESSION

COURSE ITi.E: Solid Staie Physies

COURSE CODE: PHY 314,

INSTRUCTION: Answer auy four questions. — TIME ALLOWED: 2 hrs 30mins

Ja, Of what use to modern technology is the study of solid state physics?

tb. Explain the terms: Binding energy, symmetry elements, Bravais lattices.

le. Describe the various-types. of symmetry-elements and symmetry operations present in a cubic crystal,

dd.

Show.why a 5-fold Symmetry is not permissible in a crystal lattice,

sees Was it necessary to study the thermal properties of solids?

2b. Explain Einstein's equation for molar lattice specific heat, both at lower

"Temperatures and higher temperatures.

2c At very low temperatures, the specific heat of rock salt varies with

Temperature according to Debye's. 7 law:

_ 1th, kg f TP

aan al

8 for rock salt is 281K.

How much heat is required to raise the temperature of 2kmol of rock salt from 10 to 50K?

tise" Explain the mechanism of superconductivity.

3b. What-are the properties of superconductors?

3. What are their practical uses?

ees are crystals grown?

4b.

Discuss the method of investigating the structure of a single crystal using x-rays.

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A beam of X-rays is incident on a sodium chisride crystal (lattice spacing 0.282nm). The first order Bragg reflection is observed at a glancing ;

angle of $8^\circ 35' \dots$

i What is the wavelength of X-rays?

ii At what angles would the second and third order Bragg reflections occur?

5a. Assume that the energy of two particles in the field of each other is given by the following function of the distance "r" between the centers of the particles:

$$U(r) = -\frac{24}{r^2} + \frac{2}{r^6}$$

L Show that the two particles form a stable compound for $r = 17\% a_0 = \frac{a_0}{\sqrt{7}}$

ii, Show that the potential energy of the two particles in the stable

N

configuration is equal to $-4 a$

Ca are Miller indices, illustrate with examples.

6b. What is meant by crystal imperfections?

6c. Find the angle between the directions $[2\ 1\ 1]$ and $[2\ 1\ 2]$ in a cubic crystal.

6d, The lattice constant for a unit cell of aluminum is 4.0314, Calculate the interplanar space of $(2\ 1\ 1)$ plane

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Paper 2 of 5 — PHY 314 - 2024/2025 - First Semester - Solid State Physics

Level 200 • First Semester • PastQuestion

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DEPARTMENT OF PHYSICS

HIRST SEMESTER EXAMINATION.

Course Coder PITY 314

Course Tithe Solid State Physics is

Course Unit: 3 ame 2k

Session: 2024/2025 pasts:

Answer any Four Questions

1a. Explain the following terns as applied to orystals;

Crystal Latuce

Lattice parameters of'a unit cell

Primitive cell

What is meant by symmetry elements in crystals?

1 types of symmetry elements and

\ crystal plute is mounted on air Neray speowromerer, Theglancing

5x". 12" 01", and 18° 12°, Show that these are successive

ine erystal plane, Also find the spacing, given) of X-rays in

tree reflections are

of reflection from the

0.0586 nm

Ja

The potential energy of a diatomic molecule is given in terms of the interatomic distance

r by the expression; :

$U(r) = -a/r^n$

Paper 3 of 5 — Phy 314 Exam PQ

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
First Semester Examination

Course Code: PHY314
Course Title: Solid State Physics
Course Unit: 3 units
Academic Session: 2019/2020
Time Allowed: 3hrs.

Date: 02/03/2021

Instructions: Answer any Four Questions with a minimum of one from each section. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

SECTION A

1. (a) Write short notes on the following:
(i) Crystal Structure (ii) Primitive lattice vectors (iii) Wigner – Seitz primitive cell
(b) Calculate the angle between the (111) and the (110) planes in a cubic system.
2. (a) List all the Bravais lattices and state their characteristics in terms of lattice vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, and angles α , β and γ .
(b) The lattice parameter of a crystal is 0.282nm. Calculate the angle of the second order reflection from the (111) plane for x-rays of wavelength 0.0705nm.
3. (a) Obtain the reciprocal lattice vectors for the cubic systems from the primitive translational vectors.
(b) Calculate the packing fractions for the three cubic systems.
(c) Find the Miller Indices of the planes with coordinates $(\frac{1}{4}, \frac{2}{3}, \frac{1}{2})$.
4. (a) Explain any three (3) types of symmetry operations
(b) The primitive translation vector of a two-dimensional lattice are $\vec{a} = 2\hat{i} + \hat{j}$, $\vec{b} = 2\hat{j}$, $\vec{c} = \hat{k}$. Determine the primitive translation vector of the reciprocal lattice.

SECTION B

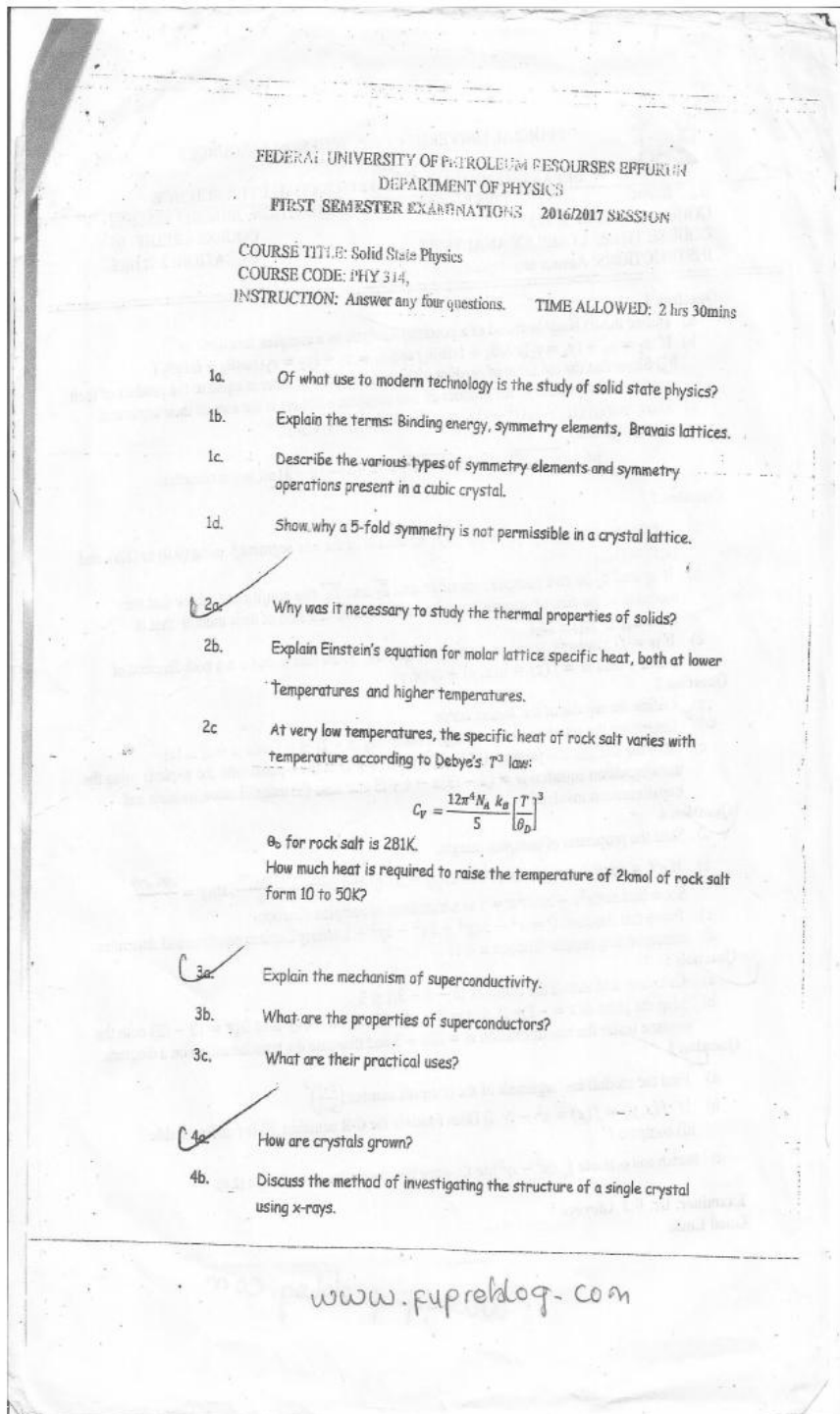
5. (ai) What is cohesive energy?
(a ii) Define the lattice energy of a solid.
(b) Explain the forces responsible for the cohesion of solids.
(c) List the different classification of solids according to their cohesion, and give an example of each classification.
6. (a) Explain the repulsive potential energy.
(b) Write down the Lennard-Jones potential energy and explain all terms in the equation.
(c) Explicitly, write down the expression for the force between the atoms ($-dU/dR$) in terms of Lennard-Jones potential energy.

Paper 4 of 5 — PHY 314 2016 2017

Level 300 • First Semester • PastQuestion

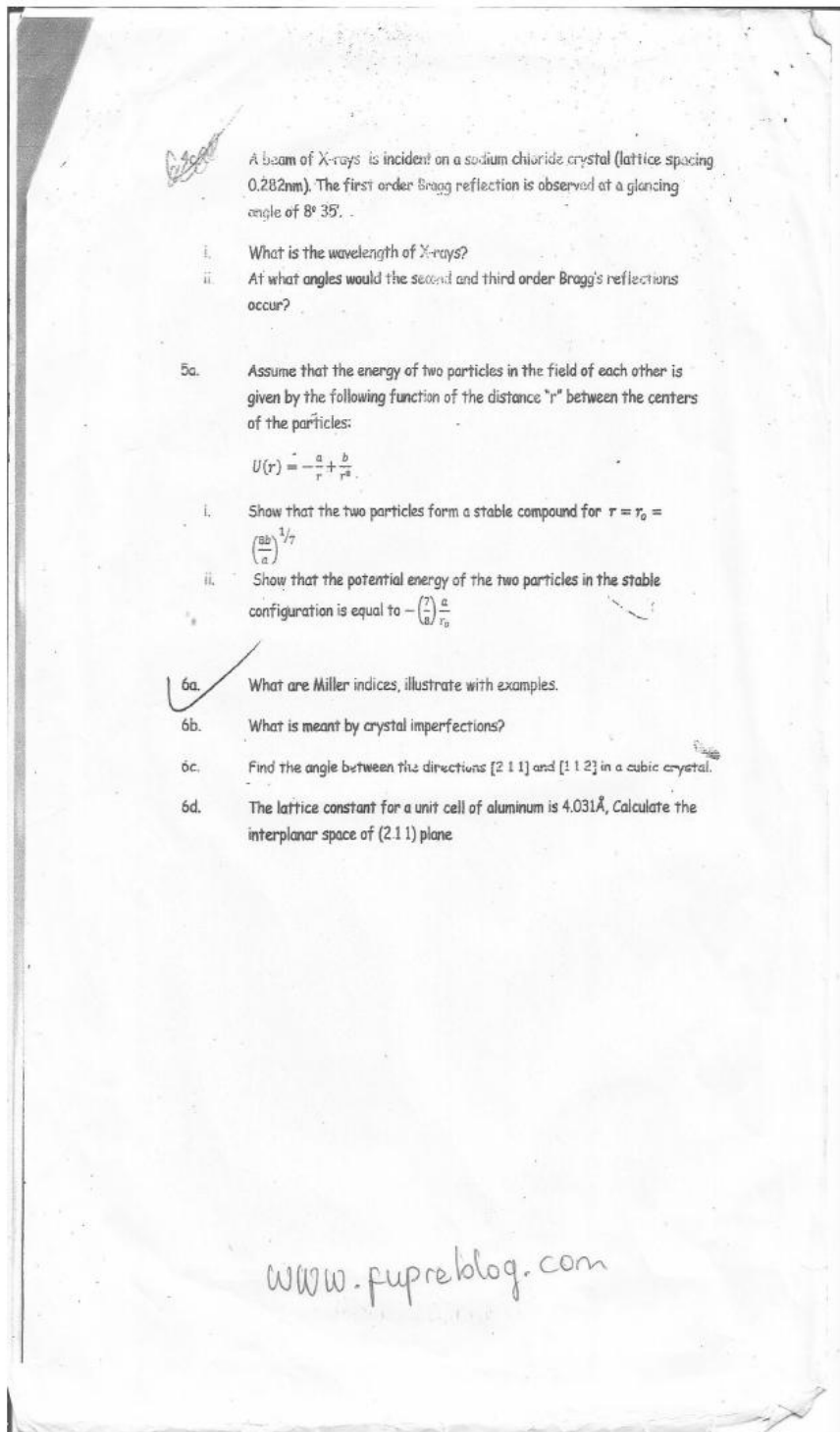
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Paper 5 of 5 — PHY 314 2017 2018 SOLID STATE PHYSICS

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATIONS 2017/2018 SESSION

COURSE TITLE: Solid State Physics
COURSE CODE: PHY 314,
INSTRUCTION: Answer any four questions. TIME ALLOWED: 2 hrs 30mins

The following expressions might be useful:

The number of energy states having energy values lying between E and E+dE is given by

$$Z(E) = \frac{\pi}{2} \left[\frac{8ma^2}{h^2} \right]^{\frac{3}{2}} E^{\frac{1}{2}} dE : \quad \vartheta_D^3 = \left(\frac{9N}{4\pi V} \right) \left[\frac{2}{\vartheta_c^3} + \frac{1}{\vartheta_l^3} \right]^{-1}$$

$$C_v = \frac{12}{5} \pi^4 R \left(\frac{T}{\theta_D} \right)^3 \text{ for } T \ll \theta_D : \quad R = 8.3143 \times 10^3 \frac{J}{KmolK}$$

1a. Discuss the salient features of Debye's theory of specific heat and show how far it agrees with the experimental values.

1b. Calculate the Debye specific heat of copper at 300K, given that the Debye characteristic frequency is $6.55 \times 10^{12} \text{ Hz}$.

1c. In aluminium, $v_l = 6.32 \times 10^3 \text{ m/s}$ and $v_t = 3.1 \times 10^3 \frac{m}{s}$. The density of aluminium is $2.7 \times 10^3 \frac{kg}{m^3}$ and its atomic weight is 26.97. Calculate Debye cut-off frequency (ν_D) for aluminium from these data.

2a. The interaction energy of two particles in the field of each other is given by $U(r) = \frac{a}{r} + \frac{b}{r^9}$ where a and b are constants
Show that the particles form a stable compound for $r = r_0 = \left[\frac{9b}{a} \right]^{\frac{1}{8}}$

2b. Show that the potential energy of the two particles in the stable configuration is $-\left[\frac{a}{r_0} \right] \left(\frac{8}{9} \right)$.

3a. Discuss a method for investigating the structure of a single crystal using x-rays.

3b. A crystal plate is mounted on an x-ray spectrometer. The glancing angles of incidence for three reflections are: $5^\circ 58'$, $12^\circ 01'$ and $18^\circ 12'$. Show that these are successive orders of reflections from the same crystal plane.

3c. Find the spacing given that λ of X-rays used is 0.0586 nm .

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- 4a. What is the 'electron gas' approximation as applied to electrical conduction in metals.
- 4b. Describe the 'free electron' model of a metal introduced by Sommerfeld.
- 4c. How would you assess the achievements of the above model, where did it fail?
- 5a. What is the significance of Miller indices?
- 5b. Obtain the Miller indices of a plane which intercepts at a , $\frac{b}{2}$, $3c$ in a simple cubic unit cell. Draw a neat diagram showing the plane.
- 5c. The Miller indices of a plane in a simple cubic crystal are (1, 2, 3), get the coordinates of the plane and sketch it.
- 6a. Define Fermi energy
- 6b. Show that at absolute zero the average energy of the electron is $\frac{3}{5}$ th of the Fermi level.
- 6c. Calculate the specific heat of a free electron gas by using classical statistics and Fermi-Dirac distribution.
- 6d. Show that the electronic specific heat of a conductor is very small compared to the lattice specific heat at room temperature.

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